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CONSTRUCTION CHEMICALS

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Anchor Grout 2K

Polyester Resin Based Anchor Grout

Description

Swadi Anchor Grout 2K is a two-component, pre-measured, polyester resin-based anchoring grout, specially formulated for rapid setting and high-strength performance. The system comprises a filled polyester resin and a hardening compound, blended to produce a high-performance resin grout suitable for both vertical and horizontal applications.

Once cured, Swadi Anchor Grout 2K exhibits excellent mechanical strength, chemical resistance, and dimensional stability. It is designed for fixing steel rebars, starter bars, threaded rods, and bolts into concrete, rock, masonry, and other substrates where the annular gap (difference between hole diameter and bar diameter) is ≤ 25 mm.

Where to Use

Swadi Anchor Grout 2K is designed for high-strength, corrosion-resistant anchoring of steel bolts, threaded rods, and reinforcement bars ranging from 12 mm to 51 mm in diameter into concrete, masonry, brickwork, and rock. It is ideal where fast installation and early load-bearing capacity are required.

Advantages

- Rapid setting and early strength gain
- Excellent corrosion resistance
- Resistant to vibrations and dynamic loads
- Shrinkage-free, non-expansive formulation
- Suitable for dry, wet, and underwater applications
- Pre-measured components for consistent mix quality
- Reliable performance in vertical and horizontal installations

Technical Data

Type	Polyester resin base
Mixing ratio	1 Part Polyster resin : 3 Part Catalyst filler
Temperature resistance	Prolonged resistance 30°c ± 70°c
Pot Life	Gelling starts approximately 10 minutes after initial mixing (at 25°C, 50% RH)
Water absorption	Nil
Specific gravity	1.9 grams/cc
Compressive Strength Test Standard BS 6319 Part 2:1983 – (50 mm x 50 mm x 50 mm cubes)	> 20 MPa (after minimum loading time) ~ 70 MPa (after 7 days)
Chemical Resistance	Resistant to fresh and salt water, petrol, oils, grease, mineral acids, and mild alkalis

Properties

Gel Temperature (°C)	Gel Temperature (°C)	Minimum Time Required Before Loading (hours)
20°C	80 min	7 hours
30°C	40 min	3 hours
40°C	15 min	1 hour

Design Criteria

Anchor Grout 2K to be used will depend upon ambient temperature and anchor conditions.

The high strength of the cured resin permits strong anchors to be created. The ultimate bond strength developed depends upon:

- Strength of host material
- Length of resin bond to bar
- Hole preparation and formation
- Type and dimension of bar

The following formula may be used to determine the minimum depth of installation for Type 1 rebar bolts, to ensure the shear stress within the concrete is kept within the limits set out in BS 8110.

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Minimum hole depth (mm) $= \frac{0.6Y}{SPd_2} \cdot \frac{Pd^2_1}{4} = \frac{0.15Yd^2_1}{Sd_2}$

Where Y is characteristic yield strength of steel (460 N/mm²)

S is permitted shear stress in concrete (N/mm2)

d_1 is bar diameter (mm)

d_2 is hole diameter (mm)

This formula is used typically as shown in Table 1.

Table 1

Minimum hole depth

Characteristic concrete strength (N/mm²):			20	25	30	>40
Bar diamete (mm)	Yield (tonnes)	Hole diameter	Minimum hole depth (mm)			
12	5.2	20	280	250	225	200
16	9.3	20	280	250	225	200
20	14.5	25	615	555	500	440
25	22.6	32	750	675	615	540
32	37.0	38	1035	930	845	745
40	57.8	45	1365	1225	1115	980

Application Instructions

Hole Preparation

- For best results, drill clean and rough-sided holes using rotary percussive drills with air or water flushing.
- Avoid smooth holes. If using diamond core drilling, under-ream the holes or apply higher safety margins.
- Cast-in holes should preferably have a reverse taper (inverted dovetail shape) for better grip. If straight holes are cast, ensure their surface is sufficiently rough to provide mechanical anchorage.

Bar Preparation

- Anchoring bars must be clean and free from oil, grease, or loose rust.
- Ideally, degrease the bars and remove flaky rust or debris using a wire brush or similar method.

Mixing

- Use entire pre-measured packs (resin and hardener-filler) – do not mix partial quantities.
- Mix thoroughly using a slow-speed mechanical mixer until a uniform, lump-free paste is achieved.
- Once mixed, use the grout within its gel time as per temperature-dependent curing table.

Installation

- Pour the mixed grout into the cleaned hole to the required depth.
- Insert the anchor bar slowly, using a gentle twisting or pushing motion to eliminate trapped air and ensure full contact with the grout.
- Leave the bar undisturbed until the grout is fully cured. Refer to the table below for curing and loading times.

Consumption Estimation

1 kg Pack (Approx. 570-600 cm³ Yield)

Figures include ~25% wastage. Suitable for standard applications. Adjust in case of deep holes, porous substrates, or old concrete.

Grout Volume Required (per 100 mm Embedment)

Hole Diameter (mm)	Bar Diameter (mm)	Approx. Grout Volume (cm³)
20	12	25
25	12	50
25	16	40
20	20	25

No. of Anchors (200 mm Deep) Fixed with 1 Kg Pack

Hole Diameter (mm)	Bar Diameter (mm)	No. of Bolts per 1 Kg Pack
20	12	11
25	12	5
25	16	7
20	20	11

Cleaning

Clean the tools using Swadi Clean Epoxy Solvent when the mixture is still fresh. The hardened mixture can be removed from tools mechanically.

Shelf Life

Swadi Anchor Grout 2K has a shelf life of 12 months when stored in normal dry conditions.

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Health & Safety

For further information, please consult the Material Safety Data Sheet (MSDS).

Condition of Sale

Sold subjects to the company's condition of sale which are available on request.

Packaging

1 kg standing pouch.

Storage and Handling Guidelines

Swadi Anchor Grout 2K should be stored in a dry, well-ventilated area, away from heat, sunlight, and moisture, and in accordance with local regulations. Keep containers tightly sealed to prevent contamination or curing, and store between 5°C and 30°C. Handle with care to avoid skin and eye contact, and always wear suitable personal protective equipment such as gloves, safety goggles, and a mask during use. Do not mix partial packs or alter proportions. Keep out of reach of children and follow usage guidelines as provided on the packaging for optimal performance.

Environmental, Health, and Safety Information

For comprehensive information regarding the safe transportation, handling, storage, and disposal of this chemical product, please refer to the accompanying Safety Data Sheet (SDS). The SDS contains detailed information about physical properties, environmental impacts, toxicological hazards, and other safety-related factors. It is crucial to read the current SDS before using any product.

Disclaimer

The information provided in this datasheet is based on our extensive experience and is given in good faith to assist you. However, we reserve the right to update this information at any time without prior notice due to our ongoing research and development efforts. While we guarantee the consistent quality of our products, we cannot be held responsible for any losses or damages that may arise from their use if site conditions or the execution of work are not properly managed. This data sheet becomes invalid upon the publication of a new edition, and please contact us for the latest version. Please note that we do not provide any warranties regarding the results of our products in specific applications, as the storage, handling, and application of materials are factors beyond our control.

Contact Information

For immediate orders or enquires, contact our **24/7 support team at +91 8792919513 or enquiry@swadigroup.co.in.** For additional support options and resources, please visit our website at swadigroup.co.in

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